Density-Aware Cloud Removal of Remote Sensing Imagery Using a Global–Local Fusion Transformer

Quan Rui, Shiyuan He, Tianyu Li, Guoqing Wang, *Member, IEEE*, Ningjuan Ruan, Lin Mei, Yang Yang, *Senior Member, IEEE*, and Heng Tao Shen, *Fellow, IEEE*

***Abstract*— Cloud cover poses a significant challenge in remote sensing image processing, affecting the extraction and analysis of terrestrial features. Despite advancements in multitemporal cloud removal methods, single-image declouding remains crucial for emergency response and disaster management, where rapid acquisition of cloud-free imagery is essential. Traditional approaches often rely on synthetic aperture radar (SAR) or cloud masks as guidance for cloud removal, introducing additional complexities and dependencies on extensive data. To address these limitations, we propose a density-aware cloud removal using a global–local fusion Transformer (DCR-GLFT), which leverages density information as guidance and does not rely on extensive data. Specifically, our method employs density labels to guide the cloud removal process through two primary stages: cloud density estimation and density-guided cloud removal. A cloud density classifier is proposed in the first stage, trained with roughly estimated ground truth, to generate density labels for guiding subsequent removal processes. The second stage integrates cloud density information with cloud–ground image features using a Transformer-based network, enabling precise and nuanced cloud removal while preserving underlying surface details through the integration of both global and local features. The proposed method achieved the state-of-the-art results (peak signal-to-noise ratio (PSNR) of 28.93 and structural similarity index measure (SSIM) of 0.84) on the renowned cloud-removal dataset SEN12MS-CR, even without utilizing SAR data for guidance. This accomplishment highlights its significant advancement in the single-image cloud removal task. Our code will be made available at https://github.com/ruiquan1214/DCR-GLFT.git.**

Received 12 July 2024; revised 30 August 2024; accepted 25 September 2024. Date of publication 10 October 2024; date of current version 12 December 2024. This work was supported in part by the National Natural Science Foundation of China under Grant U23B2011, Grant 62102069, Grant U20B2063, and Grant 62220106008; and in part by the Key Research and Development Program of Zhejiang under Grant 2024SSYS0091. *(Corresponding author: Guoqing Wang.)*

Quan Rui, Shiyuan He, Tianyu Li, and Guoqing Wang are with the Center for Future Media and the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: gqwang0420@uestc.edu.cn).

Ningjuan Ruan is with Beijing Institute of Space Mechanics and Electricity, Beijing 100094, China.

Lin Mei is with the Donghai Laboratory, Zhoushan, Zhejiang 316021, China, and also with Beijing Institute of Space Mechanics and Electricity, Beijing 100094, China (e-mail: mei_lin@vip.126.com).

Yang Yang is with the Center for Future Media and the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China, and also with the Institute of Electronic and Information Engineering, University of Electronic Science and Technology of China, Guangdong 523808, China.

Heng Tao Shen is with the School of Electronics and Information Engineering, Tongji University, Shanghai 201804, China, and also with the Peng Cheng Laboratory, Shenzhen 518066, China.

Digital Object Identifier 10.1109/TGRS.2024.3477739

***Index Terms*— Cloud removal, density label, global–local fusion Transformer (GLFT).**

I. Introduction

In THE field of remote sensing, multispectral imagery is an essential data source for analyzing surface characteristics, assessing vegetation health, monitoring land use patterns, and detecting environmental changes [1]. However, optical imagery is susceptible to cloud contamination, which obscures the view of Earth's surface and complicates terrestrial analysis [2]. According to cloud information collected by the Moderate Resolution Imaging Spectroradiometer (MODIS) on the Terra and Aqua satellite platforms, global cloud coverage is approximately 67%, with cloud cover over land areas around 55% [3], [4]. Consequently, it is essential to apply cloud removal techniques to utilize the image data effectively.

By leveraging the cloud-free portions of images captured at different times, multitemporal image cloud removal methods have achieved remarkable performance [5], [6]. However, these methods require multiple images captured at different times, which must have consistent illumination conditions and minimal landscape changes. Such methods heavily rely on temporal correlation to accurately reconstruct cloud-covered areas, which is particularly challenging in regions experiencing rapid environmental changes or where cloud-free images from different times are unavailable. Thus, there is a heightened demand for advanced cloud removal techniques specifically designed for single remote sensing image [7].

Previous research on single-image declouding has primarily focused on utilizing synthetic aperture radar (SAR) [8], [9], [10], [11], [12] or cloud-mask [11], [13] methods by incorporating additional information to facilitate cloud removal, as shown in Fig. 1. However, these approaches have inherent limitations. While SAR imaging offers valuable insights by partially penetrating clouds [14], it introduces additional complexities. The unique characteristics of SAR images require extra information integration, complicating the declouding process. Furthermore, distinguishing between surface reflection signals and cloud-induced signals demands complex algorithms, potentially introducing noise and artifacts [15], [16]. The challenges of integrating SAR data into the declouding process are further exacerbated by the data's high dimensionality and the need for specialized preprocessing steps, resulting in increased computational costs and extended

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.

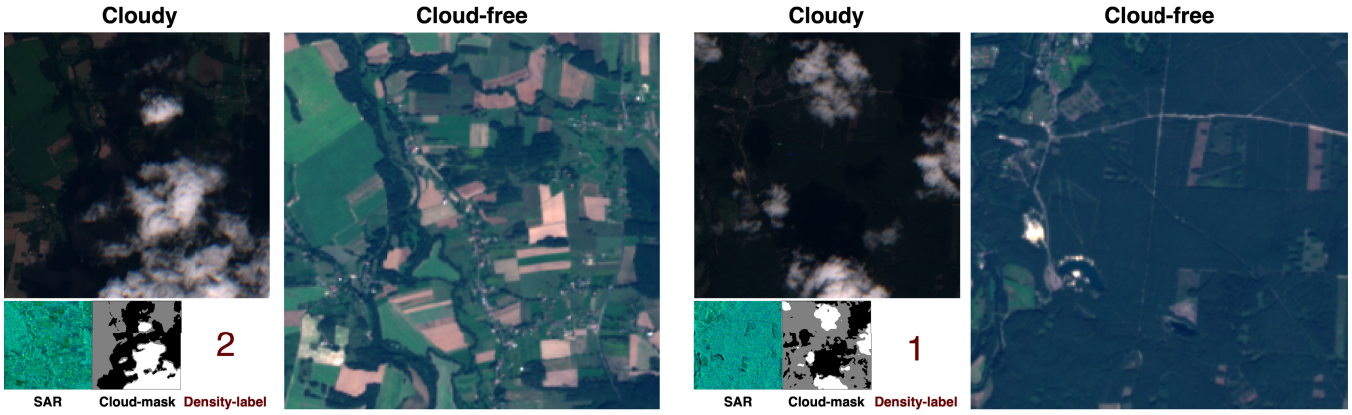


Fig. 1. Remote sensing images w/ cloud and w/o cloud, as well as methods for cloud removal tasks, are critical areas of study. Existing approaches often use SAR or cloud masks as guidance for cloud removal, introducing extra effort for obtaining more data modality or for pixel-level image labeling. In contrast, our method simply employs density labels estimated from the images themselves as guidance for the cloud removal process.

processing times. Similarly, although cloud-mask techniques have proven effective in remote sensing declouding tasks [17], [18], [19], [20], they have notable drawbacks. These methods often require additional monitoring networks for detection and removal processes. Additionally, a precise ground-truth data for high-quality cloud masks is labor-intensive and costly, as it typically involves manual curation. Furthermore, implementing these advanced algorithms can be challenging in resource-limited situations.

Therefore, we propose an innovative approach called density-aware cloud removal using a global–local fusion Transformer (DCR-GLFT). This method innovatively addresses the complexities of cloud penetration and the accurate differentiation of cloud and surface features. DCR-GLFT operates through two primary stages: cloud density classification and effective cloud removal. In the first stage, a straightforward pixel detection method is employed to determine the cloud cover levels in different images. The obtained levels are then used as ground truth to train our classification network, and the predicted density level information is utilized as guidance to improve the accuracy and efficiency of the cloud removal task. In the second stage, a Transformer-based operation is utilized, which integrates the cloud density information derived in the first stage with the cloud–ground image features. This integration facilitates a targeted and nuanced cloud removal process, allowing for the preservation of underlying surface details and minimizing the introduction of artifacts and noise. The fusion approach of the Transformer leverages both global and local image features, ensuring a comprehensive and precise declouding outcome.

In summary, this article makes the following contributions.

1) We introduce a cloud density classifier that, through an initial estimation of cloud cover, guides the declouding model in learning the spectral relationships of multispectral imagery, thereby enhancing cloud removal performance.
2) Our method employs a GLFT (DCR-GLFT) to integrate cloud density information with cloud–ground image features. This approach facilitates more detailed and precise cloud removal, ensuring the integrity of underlying surface details while minimizing the introduction of noise and artifacts.
3) Our method demonstrated the state-of-the-art performance on the prestigious SEN12MS-CR cloud removal dataset, achieving a peak signal-to-noise ratio (PSNR) of 28.93 and an structural similarity index measure (SSIM) of 0.84 even without utilizing SAR data for guidance. This achievement underscores its significant advancement in the domain of single-image cloud removal.

II. Related Work

Cloud detection is typically a critical step in the process of image declouding and restoration. By treating it as a pixelwise binary classification problem, cloud detection involves classifying each pixel in an image as either cloud or noncloud. Various methods have been developed to detect clouds and cloud shadows, which can be classified into temporal and nontemporal approaches [21], as well as nondeep learning and deep learning techniques [22], [23]. With the development of convolutional neural networks, many researchers have proposed effective deep learning models for cloud detection, such as stacked autoencoders (SAEs) [15] and DeepLabV3+ based on deep convolutional networks [24], as well as more recent models like DABNet [19] and Cloud-Net [20], which are better suited for image processing tasks. These methods are capable of generating high-quality mask files but often at the cost of neglecting the information about the surface underneath the clouds [22], [23]. To address the issue of inaccurate mask files leading to unsatisfactory cloud removal results, Lin et al. [16] utilized the RTCR method and augmented Lagrange multipliers. Typically, the signals received by remote sensing imaging sensors are a superposition of surface reflection signals and cloud reflection signals [18], [25]. Simple classification methods can only locate and identify clouds in images but fail to estimate cloud coverage or recover surface information. Recognizing that clouds in images are usually mixed with surface information and different transparency levels lead to different superposition patterns, Li et al. [17] proposed a hybrid cloud detection algorithm by fully leveraging various algorithms to maximize detection accuracy. However, the performance of these deep learning-based methods heavily relies

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.

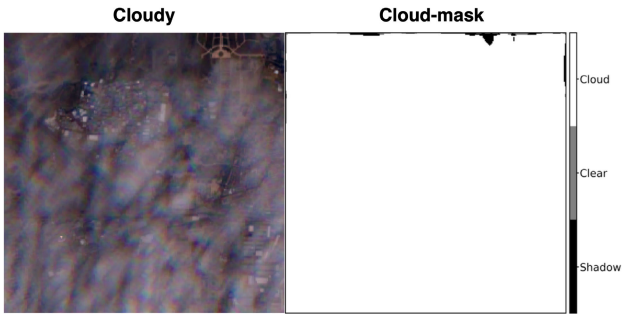


Fig. 2. Image with clouds and the cloud mask obtained using the DSen2-CR method [11]. It can be seen that for thin clouds, it is difficult to delineate the relationship between the cloud layer and the ground scene.

on manually labeled training data, which is generated through a tedious and labor-intensive process. Moreover, clouds in remote sensing images are often intertwined with ground surface information, as shown in Fig. 2, and varying levels of transparency can complicate the labeling process. As a result, accurately predicting the cloud mask is challenging, and inaccuracies in the predicted cloud mask can adversely affect the cloud removal process.

Cloud removal aims to reconstruct missing information in optical satellite imagery caused by clouds. The general development process started with early methods that predicted missing data by assuming similar statistical and geometrical structures between the damaged and remaining regions [26], [27]. Subsequent research leveraged the generative capabilities of generative adversarial networks (GANs) to learn mappings between cloudy and cloud-free images [28], [29], [30]. Multitemporal methods further advanced the field by using data from other time periods to restore missing information [31], [32], [33], [34], [35]. Specifically, early approaches treated cloud detection and removal tasks separately [18], [25]. Many researchers use deep learning techniques for single-image cloud removal. The widely used dark channel method has excellent mathematical derivations [19]. However, its applicability may be limited due to differences in imaging between satellite images and other types of images. Moreover, there are some errors in transmittance estimations and the dark channel prior, so images are prone to dimming or even distortion after cloud removal. The k-nearest neighbor (KNN) matting method falls under nonlocal matting. It assumes that the transparency of a pixel can be described by weighting the transparency values of nonlocal pixels with a similar appearance, such as matching the color and texture. The goal is to allow the transparency value to propagate in nonlocal pixels. This includes laborious computations due to the comparison with nonlocal images. KNN matting improves nonlocal matting by only considering the first K neighbors in the high-dimensional feature space. It reduces the amount of computation by only considering similarities between the color and the position in their feature space. The drawback of this method is that it requires a priori trimap as input and usually leaks pixels. Defining a general feature space with few parameters is difficult. Closed-form matting [36] assumes that the reflectivity of the foreground and background is the same in the local range of the sliding window and solves the transmittance formula using the color-line model and the ridge regression optimization algorithm. However, the clouds are easily overcorrected, and the solution requires an accurate trilateral as an a priori input, which significantly limits the application of closed extinction methods. The conditional generation countermeasure network (CGAN) [37] can reconstruct damaged information well when entities are still visible. However, the number of objects in remote sensing images has greatly increased. Therefore, the generative countermeasure network exhibits noticeable distortions in the thick cloud area. Isola et al. [38] proposed an image-to-image translation method (Pix2pix) based on CGAN to achieve image-to-image generation, providing a new method for image declouding restoration. Ramjyothi and Goswami [39] used GAN to repair the ground cover information under clouds in remote sensing images. Pan et al. [18] and Emami et al. [40] introduced spatial attention to GAN to control the redundancy of the model. Wen et al. [41] used a residual channel attention network for cloud removal. Through the solid-fitting ability of deep learning, the models can effectively learn the difference between the features of clouded and cloudless images, and then, they can directly restore the absolute brightness value of the surface using image reconstruction. Cloud removal based on GANs for reconstructing surface information is one of the trending research topics in recent years. However, the biggest drawback of deep learning is that "it cannot admit that it does not know when thin and thick clouds coexist" [42]. Although the models achieve high-performance metrics, there remains a significant discrepancy between the generated images and the real images. Limited by the potential variability introduced by these methods, we use a feature extraction network for both detection and removal tasks and utilize the density label obtained from detection to guide the Transformer in learning the interchannel relationships under varying levels of cloud cover, thereby effectively avoiding the redundancy associated with two separate processes.

III. Method

A. Problem Statement

Given a cloud-covered image J, defined over the space $\mathcal{Y} \triangleq \mathcal{D} + \mathcal{P}$, where $\mathcal{D}$ and $\mathcal{P}$ denote the cloud-covered and cloud-free regions, respectively, the objective of cloud removal is to restore the cloud-covered segment $J_{\mathcal{D}}$ of the image. This task is inherently challenging due to the significant information loss caused by cloud occlusion in optical satellite observations.

The traditional strategy for cloud removal is akin to an inpainting task, where the goal is to reconstruct the cloud-obscured region $J_{\mathcal{D}}$ using the observable part $J_{\mathcal{P}}$

$$J_{\mathcal{D}} = \mathbf{G}_{\text{INP}}(J_{\mathcal{P}}; T(J)) \tag{1}$$

where $\mathbf{G}_{\text{INP}}$ is an inpainting operator guided by the latent structures of the entire image, represented as $T(J)$. These structures may include image priors such as smoothness and nonlocal similarities, or features derived from data, facilitating the restoration process. However, for varying cloud densities, these latent structures can be incomplete or absent to varying degrees, leading to potential reconstruction failures.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.

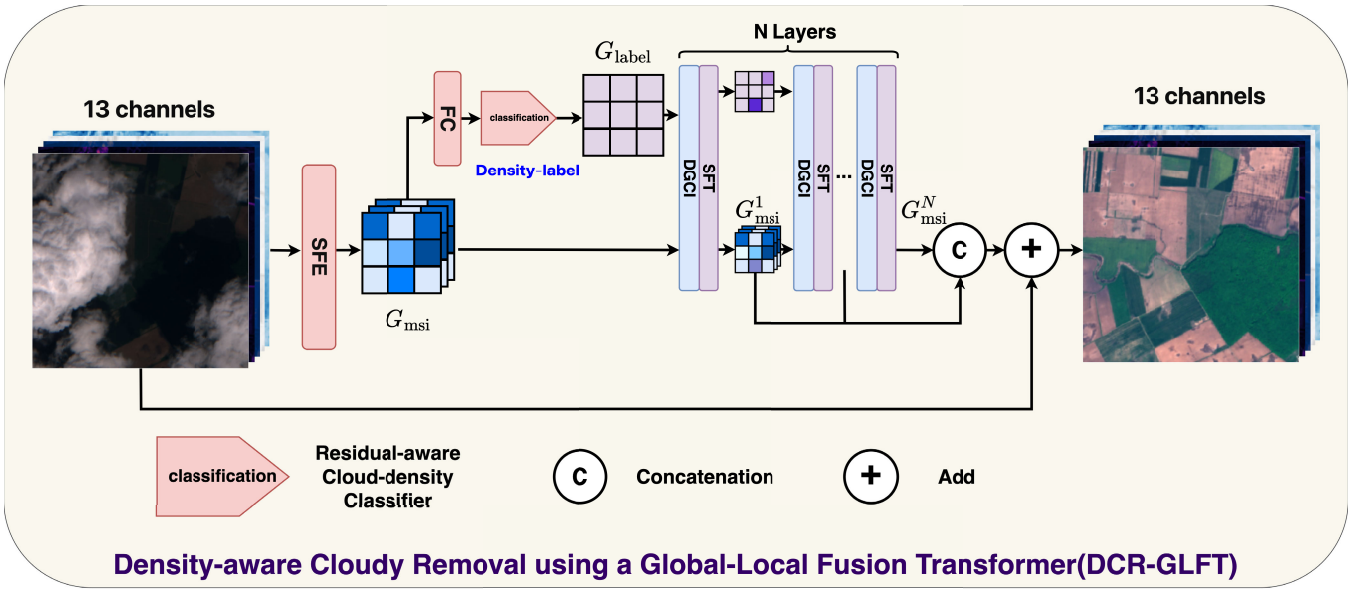


Fig. 3. Overview of the proposed DCR-GLFT algorithm. The algorithm consists of two parts: the first part performs cloud coverage classification using an auxiliary-guided cloud classifier, and the second part uses a DGCI block to guide the learning of relationships among local multispectral windows based on the obtained density-label information. Additionally, the SFT block is employed to convey the cloud coverage information carried by the density label.

In contrast to methods relying on external data, our density-label approach simplifies the cloud removal process by directly utilizing inherent cloud density information

$$J_{\mathcal{D}} = \mathbf{G}_{\text{DLD}}(J_{\mathcal{P}}; Z(J)) \tag{2}$$

where $\mathbf{G}_{\text{DLD}}$ is a declouding operator conditioned on $Z(J)$, the cloud density information. $Z(J)$ provides essential guidance by classifying cloud densities, thus enabling a targeted and nuanced removal process.

Our method efficiently utilizes cloud density information as guiding information to direct the interchannel relationships in multispectral images, reconstructing cloud-covered areas with high fidelity. By analyzing $Z(J)$, it guides the specific handling of different levels of cloud density, thereby optimizing cloud removal performance.

To address this, we have implemented a density-aware fusion strategy, developing an operator $\mathbf{G}_{\text{DCR-GLFT}}$ that integrates both global image structures and localized density-specific information

$$J_{\mathcal{D}} = \mathbf{G}_{\text{DCR-GLFT}}(J_{\mathcal{P}}, Z(J); T(J)) \tag{3}$$

where $T(J)$ not only includes standard image priors but also adapts to the nuanced interactions dictated by the cloud density labels $Z(J)$. This approach ensures that the cloud removal process is more precise and effective for different cloud densities.

B. Overview

Our proposed method, termed DCR-GLFT, primarily consists of three components: the auxiliary-guided cloud classifier, the density-aware global context interaction (DGCI), and the spatial feature transform (SFT), as shown in Fig. 3. The density label guides the multispectral features, enabling the model to learn relevant content based on varying levels of cloud coverage. Specifically, the input cloudy image J is processed for feature extraction through a shallow feature extraction (SFE) module, which extracts modality-specific features

$$G_{\text{msi}} = \text{SFE}_{\text{msi}}(J) \tag{4}$$

where SFE_{msi} represents the shallow feature extraction function applied to image J.

In the auxiliary-guided cloud classifier block, the obtained G_{msi} is flattened and input into a classifier to predict the cloud density level. The predicted logits are converted into class indices, representing the cloud density categories. These indices are expanded to match the spatial dimensions of the input image, ensuring consistency for subsequent processing, resulting in G_{label}.

In the DGCI block, G_{msi} and G_{label} are input into a series of N DGCI blocks and STF blocks to obtain comprehensive knowledge features. The intermediate features $G^1_{\text{msi}}, G^2_{\text{msi}}, \ldots, G^N_{\text{msi}}$ are obtained as follows:

$$G^i_{\text{msi}} = \text{STF}\big(\text{DGCI}\big(G^{i-1}_{\text{msi}}, G_{\text{label}}\big)\big). \tag{5}$$

Finally, by aggregating all intermediate multispectral features, a high-quality cloud-free image J_D is reconstructed

$$J_{\mathcal{D}} = J + \text{IR}\big(\big[G^1_{\text{msi}}, G^2_{\text{msi}}, \ldots, G^N_{\text{msi}}\big]\big) \tag{6}$$

where IR denotes the function of cloud-free image reconstruction, which upsamples the feature maps and uses convolution layers to convert the feature maps to the target number of channels.

C. Auxiliary-Guided Cloud Classifier

As previously discussed, although some earlier methods have significantly improved cloud removal performance, they typically incorporate a cloud mask as a cloud density label to guide the declouding process. However, these methods face several challenges: first, manually annotating cloud masks is costly and the available data are limited; second, these

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.

methods often rely on complex networks to process and generate cloud masks, which increases the difficulty of implementation. Instead of predicting the whole mask, we found that integrating cloud density information into the network can enhance the overall learning process and thereby ensure better generalization under different cloud conditions in our experiments. To estimate cloud density from given input images, we propose a density-aware cloud density classifier that uses density labels to represent cloud features, avoiding the difficulties of obtaining cloud masks. Moreover, to train this classifier, we generated density labels for cloud-containing images in the training dataset using the method described by Meraner et al. [11] and analyzed the impact of various labels on the results in Section IV. Meraner et al.'s [11] method involves extracting clouds and their shadows by comparing pixel values. Specifically, their approach detects clouds by analyzing the brightness values and spectral features of the pixels, generating an initial cloud mask. Further comparison of pixel values is then used to determine the cloud density labels. To reduce computational complexity, we do not precisely delineate clouds and shadows to obtain our ground-truth labels but rather generate cloud coverage levels. These cloud coverage levels are obtained using different cloud coverage amounts as described in Meraner et al.'s [11] method. This approach reduces computational complexity while effectively representing cloud coverage. To simplify the model, we used the same feature extractor as discussed in Section III-B for extracting classification features, and the classifier consists of a fully connected layer. The final layer contains a set of n neurons that indicate the cloud density category of the input image. The ablation study discussed in Section IV-B demonstrates the effect of different numbers of categories on model efficacy.

D. Density-Aware Global Context Interaction

The DGCI block plays a pivotal role by integrating both local and global features to enhance cloud removal performance. The details of the DGCI block are shown in Fig. 4. This block is configured with two parallel streams for processing input multispectral images with cloud coverage and density-label features. The block follows the SAR-guided global context interaction (SGCI) proposed by Xu et al. [12] and extracts rich local features through densely connected convolutional layers. Although the density-label does not contain detailed texture information, effectively integrating it with multispectral cloud-covered images is essential, as discussed in Section IV-D. A Swin Transformer layer (STL) [43] is added after each local convolutional layer for cross-window feature interaction.

This network employs a dense connection strategy similar to the residual dense block (RDB), which is crucial for effectively extracting rich local features under various cloud densities. Unlike previous models, our model uses the expanded density labels M_{label} to guide the information flow between different channels of multispectral remote sensing images. Specifically, through the use of density labels, the model can focus on learning the weights of different channels, thereby utilizing the information from cloud-free channels to compensate

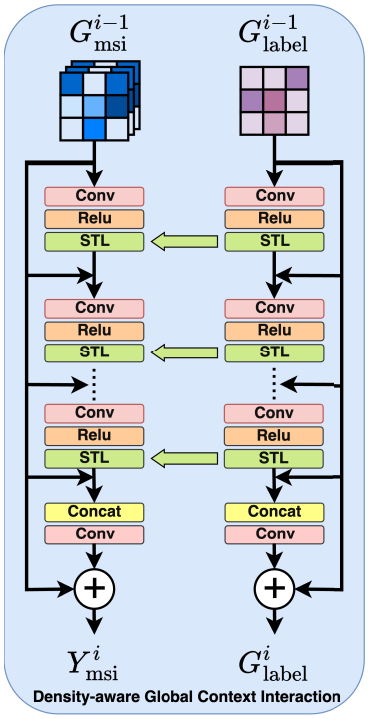


Fig. 4. Detail of the DGCI block. STL denotes the Swin Transformer layer.

for the missing information in cloud-covered regions. This approach allows the model to handle cloud-covered areas more accurately, enhancing the overall integrity and detail restoration of the image. To facilitate robust global and local feature interactions, inspired by the Transformer's capability to efficiently propagate information, we integrate an STL after each convolutional layer. The STL initially segments the input features into nonoverlapping $M \times M$ windows and computes the self-attention mechanism independently for each window.

Specifically, the transformed features of the multispectral cloud-covered image, $G_{\text{msi}} \in \mathbb{R}^{M^2 \times C}$, are linearly transformed into queries $Q_{\text{msi}} \in \mathbb{R}^{M^2 \times d}$, keys $K_{\text{msi}} \in \mathbb{R}^{M^2 \times d}$, and values $V_{\text{msi}} \in \mathbb{R}^{M^2 \times d}$, where d is the dimension of the keys or queries. Apply the same processing to G_{label}. The self-attention weights are computed as

$$M_{\text{msi}} = \frac{Q_{\text{msi}} K^{\top}_{\text{msi}}}{\sqrt{d}} + B, \quad M_{\text{label}} = \frac{Q_{\text{label}} K^{\top}_{\text{label}}}{\sqrt{d}} + B \tag{7}$$

where B represents the learnable relative positional encoding. Additionally, we introduce the density label map M_{label}, using it to guide interchannel relationships to facilitate interactions between cloud-dense and cloud-free areas.

We utilize M_{label} to refine the attention map of the multispectral image M_{msi}. The refined attention map $\hat{M}_{\text{msi}}$ is then adjusted by the gating function

$$\hat{M}_{\text{msi}} = M_{\text{msi}} + (M_{\text{msi}} - M_{\text{label}}) \odot G(M_{\text{label}}) \tag{8}$$

where $G(\cdot)$ is the gating function that modulates attention weights, and $\odot$ denotes the elementwise multiplication.

Finally, for the ith module, the output for the multispectral features is computed as

$$Y^{i}_{\text{msi}} = \text{Softmax}\left(\hat{M}^{i}_{\text{msi}}\right) V^{i}_{\text{msi}}. \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.

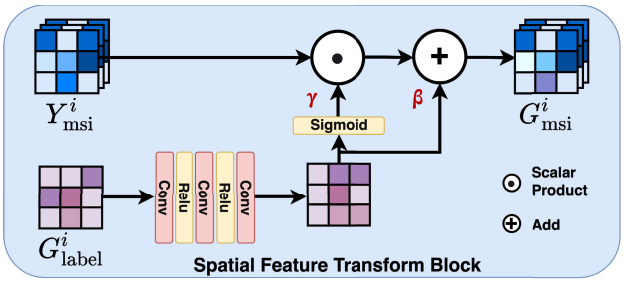


Fig. 5. Structure of the SFT block. i denotes the ith SFT module.

This methodology, referred to as global fusion, utilizes the density label M_{label} to guide the relationship among all local window features of the multispectral cloud-covered images, significantly enhancing the precision and effectiveness of cloud removal in multispectral imagery.

E. Spatial Feature Transform

We introduce a new SFT block, as shown in Fig. 5, specifically designed to modulate features in multispectral images covered by clouds based on density labels. This layer leverages density label information to guide interchannel feature learning, thereby computing modulation parameters. These modulation parameters are used to adjust the features of the input image, enabling the model to better handle cloud-covered areas. Specifically, density labels provide guiding features that help the model focus on learning the weights of different channels, allowing precise modulation and optimization of features in multispectral images.

The fundamental role of the SFT is to adaptively influence the outputs by applying an affine transformation to each intermediate feature map through calculated modulation parameters.

Initially, the density labels are processed through a sequence of convolutional layers to generate pairs of modulation parameters, specifically scale and shift parameters

$$g = \text{Conv3}\big(\text{ReLU}\big(\text{Conv2}\big(\text{ReLU}\big(\text{Conv1}\big(Y^i_{\text{label}}\big)\big)\big)\big)\big) \tag{10}$$

where Y^i_{label} represents the input density-label features. The convolution layers Conv1, Conv2, and Conv3 successively process these features to generate the final modulation parameters.

The scale parameters (γ) and shift parameters (β) are derived from the processed guide

$$(\gamma, \beta) = \text{split}\big(g, Y^i_{\text{msi}}\big). \tag{11}$$

γ is then processed through a Sigmoid activation function to ensure that its values are positive, enabling proper amplification of image features

$$\gamma = \text{Sigmoid}(\gamma). \tag{12}$$

Finally, the scaling and shifting parameters are applied to the input features to achieve feature modulation

$$G^i_{\text{msi}} = Y^i_{\text{msi}} \odot \gamma + \beta \tag{13}$$

where $\odot$ denotes the elementwise multiplication, i.e., the Hadamard product.

This approach allows the SFT to effectively adjust the features of multispectral images based on density labels extracted from cloud-covered imagery, which is crucial for enhancing the quality and effectiveness of cloud image processing tasks.

F. Loss Function

To effectively train the cloud removal classifier, we employ a dual-stage training approach. Initially, a feature extraction network is trained to capture the relevant features of cloud-covered images. Subsequently, a classification subnetwork is trained using these extracted features as input and optimized using the obtained rough cloud density labels. Finally, the two phases—feature extraction and classification—are optimized jointly. The total loss function used to train the cloud-aware classifier is as follows:

$$L_{\text{class}} = L_C \tag{14}$$

where L_C represents the cross-entropy loss for cloud-density classification.

We define a generator loss L_{gen} aimed at improving the fidelity of the cloud-free image reconstruction

$$L_{\text{gen}} = L1(J_{\mathcal{D}}, J_{\mathcal{DT}}) \tag{15}$$

where $J_{\mathcal{D}}$ and $J_{\mathcal{DT}}$ represent the predicted cloud-free images and the actual cloud-free images, respectively. This L1 loss function encourages the model to produce predictions that closely match the ground truth, effectively minimizing the absolute differences between them.

To balance the training between accurate cloud classification and effective cloud-free image reconstruction, we employ a dynamic weighting mechanism to integrate L_{gen} and L_{class} into a total loss L_{total}

$$L_{\text{total}} = \text{weight}_G \cdot L_{\text{gen}} + \text{weight}_{\text{classify}} \cdot L_{\text{class}} \tag{16}$$

where the weights weight_G and $\text{weight}_{\text{classify}}$ are determined dynamically based on the ratio of the magnitudes of L_{gen} to L_{class}

$$\text{weight}_G = \frac{\text{ratio}}{\text{ratio}+1}, \quad \text{weight}_{\text{classify}} = \frac{1}{\text{ratio}+1} \tag{17}$$

with $\text{ratio} = (L_{\text{gen}}/(L_{\text{class}} + 1e-6))$, where $1e-6$ to prevent division by zero.

This approach not only ensures optimal balance in training focus but also enhances the overall performance and reliability of the cloud removal process, effectively addressing the complexities associated with different cloud densities.

IV. Experiment

A. Dataset

The SEN12MS-CR dataset is a cutting-edge, multimodal, and mono-temporal dataset derived from the European Space Agency's Copernicus mission satellites, Sentinel-1 and Sentinel-2, specifically designed for cloud removal in Earth observation. This dataset includes multispectral optical satellite observations that encompass both clouded and clear conditions. The multispectral data consist of multiple

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE COMPARISONS OF THE PROPOSED DCR-GLFT AND THE STATE-OF-THE-ART METHODS

Method	Input				PSNR↑	SSIM↑	SAM↓	MAE↓
	Cloudy	SAR	Cloud-mask	Density-label				
SAR-Opt-cGAN(Grohnfeldt et al., 2018)	✓	✓			25.59	0.764	15.494	0.043
SAR2OPT(Bermudez et al., 2018)		✓			25.87	0.793	14.788	0.042
Simulation-Fusion GAN(Gao et al., 2020)	✓	✓	✓		24.73	0.701	16.633	0.045
SpA GAN(Pan, 2020)	✓				24.78	0.754	18.085	0.045
DSen2-CR(Meraner et al., 2020)	✓	✓	✓		27.76	0.874	9.472	0.031
GLF-CR(Xu et al., 2022)	✓	✓			28.64	**0.885**	8.981	0.028
UnCRtainTS L2(Ebel et al., 2023)	✓				28.90	0.880	8.320	**0.027**
DCR-GLFT(Ours)	✓			✓	**28.93**	0.838	**7.948**	**0.027**

spectral channels, such as visible, near-infrared, and short-wave infrared bands. These multispectral observations capture rich spectral information, allowing for effective differentiation between cloud-covered and cloud-free areas, thereby facilitating high-precision feature extraction and analysis. As the first public dataset providing expansive, global coverage for all seasons, SEN12MS-CR facilitates the advancement of contemporary machine learning techniques aimed at cloud removal. The dataset includes observations from 175 globally dispersed regions of interest (ROIs) across all four seasons of 2018, structured into 122 218 patch triplets. Each triplet is meticulously sized and co-registered, ensuring seamless integration with the SEN12MS dataset. With a cloud coverage of around 48%, SEN12MS-CR spans a wide range of atmospheric conditions, from unobstructed clear skies to heavy cloud cover, establishing it as an essential tool for developing and testing new automated cloud removal technologies.

B. Implementation Details

We randomly divided 175 ROIs into 165 for training, ten for validation, and ten for testing. To prevent the overall performance from being biased toward any specific season, we allocated images from four different seasons to both the validation and test sets. Specifically, the training, validation, and test sets consist of 107 816, 7655, and 6747 samples, respectively. The results of the cloud removal are evaluated using normalized data based on the PSNR, SSIM, spectral angle mapper (SAM), and mean absolute error (MAE).

To accurately label different levels of cloud cover within our dataset, we utilize the cloud detection flowchart presented by previous researchers. Each patch is then labeled according to the cloud coverage calculated using the aforementioned method. We categorize cloud cover into three distinct labels: low, medium, and high. These categories are defined by the following ranges of cloud coverage: 0%–33%, 33%–67%, and 67%–100%. The cloud coverage range includes clouds and their associated shadows.

C. Comparisons With the State-of-the-Art Methods

We evaluate the DCR-GLFT model against a variety of leading cloud removal technologies. These include spatial attention-based SpA GAN [8], SAR-to-optical conversion strategies like SAR2OPT [9], and methods that integrate SAR and optical data such as SAR-Opt-cGAN [10], Simulation-Fusion GAN [13], DSen2-CR [11], GLF-CR [12], and UnCRtainTS L2 [44].

SpA GAN, for instance, processes all channels from an input optical image, employing a spatial attention network (SPANet) [45] as its generator to transform cloudy images into clear ones. SAR2OPT uses the U-Net architecture to facilitate SAR to optical image translation without depending on additional optical satellite data. Both SAR-Opt-cGAN and DSen2-CR utilize SAR imagery as a preliminary guide to assist in reconstructing scenes obscured by dense clouds. These methods concatenate SAR data directly with optical image channels to simulate the complete spectrum of optical bands. SAR-Opt-cGAN builds upon the U-Net framework, whereas DSen2-CR expands upon the EDSR architecture [46]. Simulation-Fusion GAN initially converts SAR data into simulated optical formats before merging these with real SAR and degraded optical images to enhance prediction accuracy. GLF-CR utilizes Transformer networks to extract features from SAR images to assist in cloud removal for cloud-covered images [12]. UnCRtainTS leverages temporal information from time-series data to improve the clarity and quality of the reconstructed images, making it a valuable tool for remote sensing applications where cloud cover often poses significant challenges [44].

To verify the superiority of the DCR-GLFT model utilizing density labels, our quantitative results are presented in Table I, where the SAM is expressed in degrees. Compared to other state-of-the-art methods, the proposed DCR-GLFT network demonstrates significant improvements. The results for SpA GAN, DSen2-CR, Simulation-Fusion GAN, SAR2OPT, SAR-Opt-cGAN, GLF-CR, and UnCRtainTS L2 were all obtained from the SEN12MS-CR official website. To validate the superiority of DCR-GLFT in utilizing density labels, we tested using the pretrained models provided by SpA-GAN and DSen2-CR. The quantitative results are presented in Table I. We selected two scenes to evaluate the qualitative results, as shown in Fig. 6. For each scene, from top left to bottom right, the images are, respectively, the cloudy image, SpA GAN, DSen2-CR, DCR-GLFT, cloud-free image, cloud-free (enlarged) image, SpA GAN (enlarged), DSen2-CR (enlarged), DCR-GLFT (enlarged), and cloud-free (enlarged) image.

Compared to SpA-GAN and DSen2-CR, it can be observed that DSen2-CR performs better in removing cloud shadows but

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.

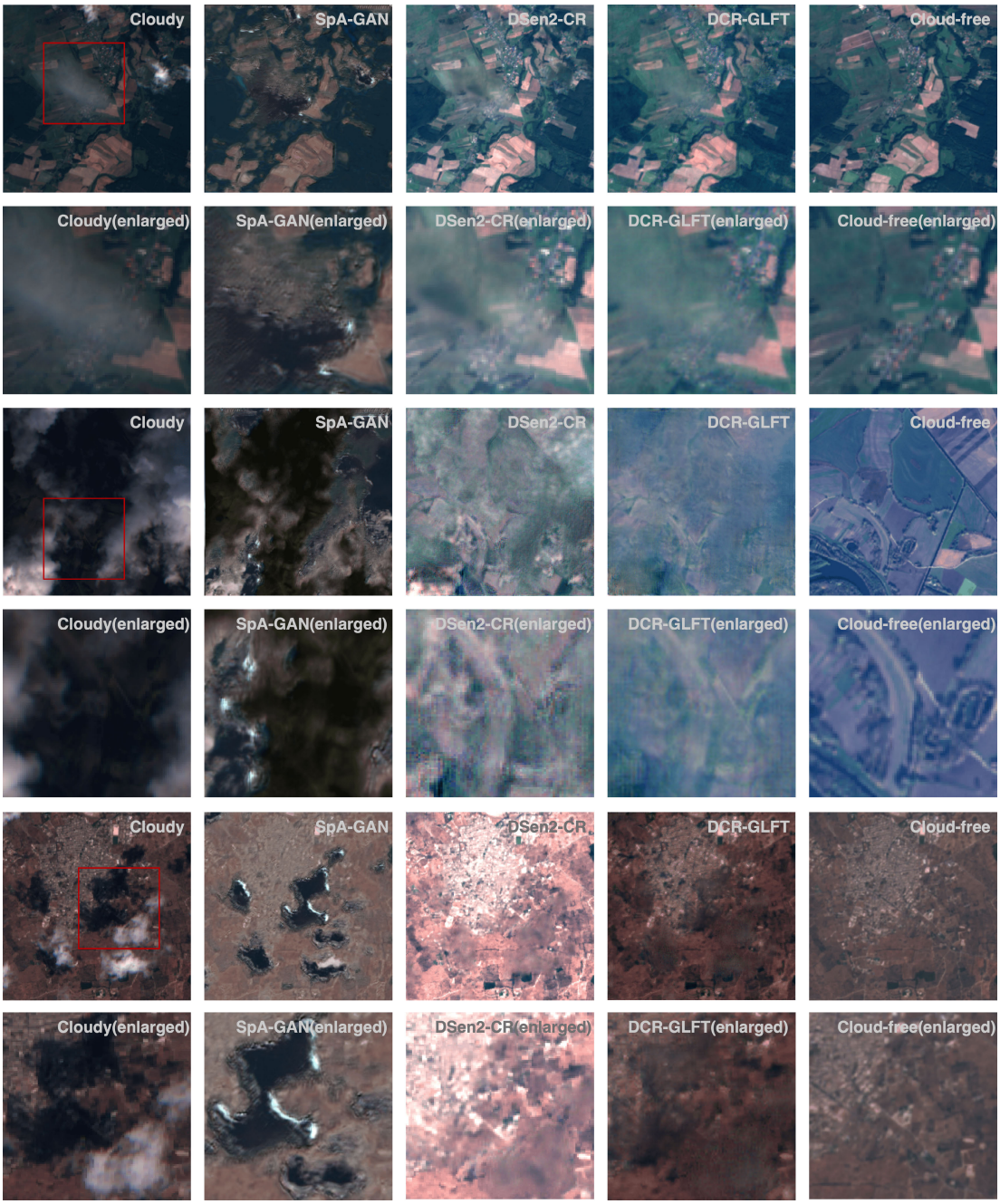


Fig. 6. Qualitative results of cloud removal for two different scenes. For each scene, from top left to bottom right, the images are, respectively, the cloudy image, SpA GAN, DSen2-CR, DCR-GLFT, cloud-free image, cloud-free (enlarged) image, SpA GAN (enlarged), DSen2-CR (enlarged), DCR-GLFT (enlarged), and cloud-free (enlarged) image. The term "enlarged" indicates a magnified view of the region marked by the red box in the cloudy image.

exhibits weaker performance in preserving fine details. It often blurs the boundaries between different scenes and leaves a noticeable smearing effect after cloud removal. As shown in the first scene of Fig. 6, the removal of small clouds by DSen2-CR results in significant color distortion, with clouds being more interpreted as ground information. SpA-GAN shows inadequate performance in handling thick clouds and fails to effectively remove them. Conversely, DCR-GLFT successfully restores certain details in cloud-covered areas and excels in preserving the boundaries of different terrains. As shown in the second scene of Fig. 6, even for images with substantial cloud cover, DCR-GLFT can effectively restore some of the information.

D. Grading of Different Cloud Covers

To assess the impact of varying numbers of cloud density labels on the task of cloud removal, we experimented with three different categorizations of cloud coverage, specifically configuring sets with three, four, and five distinct cloud cover levels. The quantitative results are presented in Table II. These classifications were derived using the cloud detection flowchart mentioned in [11]. The categories are defined by the following ranges of cloud coverage:

1) *for the three levels:* 0%–33%, 33%–67%, and 67%–100%;
2) *for the four levels:* 0%–25%, 25%–50%, 50%–75%, and 75%–100%.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTITATIVE ABLATION STUDY ON DIFFERENT DENSITY-LABEL CLASSIFICATIONS

Method	PSNR↑	SSIM↑	SAM↓	MAE↓
3 levels	**28.93**	**0.838**	**7.948**	**0.027**
4 levels	28.70	0.867	7.947	0.028
5 levels	28.39	0.847	8.305	0.029

TABLE III
QUANTITATIVE ABLATION STUDY OF DIFFERENT MODULES

Method	PSNR↑	SSIM↑	SAM↓	MAE↓
Baseline	24.98	0.589	12.868	0.046
Baseline w/ DL	25.02	0.602	12.370	0.046
DGCI w/o DL	28.08	0.875	8.531	0.030
DGCI w/o SFT	28.39	0.847	8.305	0.029
DCR-GLFT	**28.93**	**0.838**	**7.948**	**0.027**

3) *for the five levels:* 0%–20%, 20%–40%, 40%–60%, 60%–80%, and 80%–100%.

The outcomes of these variations are systematically tabulated and analyzed to determine the efficacy of each classification in improving the accuracy and efficiency of cloud removal operations. The results are presented in Table II, illustrating the relative performance enhancements across different configurations.

The analysis conclusively shows that a three-level cloud density labeling system provides an optimal balance between model complexity and the ability to effectively perform cloud removal tasks. By reducing the granularity of the categories, it encompasses a more balanced distribution of cloud densities, preventing the model from overfitting to specific, narrow cloud density ranges. This balance facilitates stronger generalization across different cloud conditions, allowing the model to focus more on significant differences that affect the overall visibility and quality of the image, thereby enhancing cloud removal performance.

E. Ablation Study

To validate the effectiveness of the modules in the proposed DCR-GLFT, we analyzed the efficacy of each component by comparing variants with and without the density label, STL layer (STL), and STF block (STF). The qualitative and quantitative results are displayed in Table III and Fig. 7. From Table III and Fig. 7, we can draw the following conclusions.

1) Baseline: We trained a network with fully connected layers as the baseline to demonstrate the importance of each module for the task of cloud removal. The network processes the input through two convolutional layers, each followed by max pooling to reduce spatial dimensions while capturing essential features. These layers help discern textures and structures obscured by clouds. The output from these layers is then flattened and passed through a dense network comprising two fully connected layers. The first fully connected layer transforms the condensed features into a 512-D space, which not only serves to capture deeper feature interactions but also feeds into a subsequent layer that reconstructs the image to its original dimensions. From Fig. 7, it can be seen that although the fully connected layers can generate some information, their grasp of details is very poor, and the output contains a lot of noise.

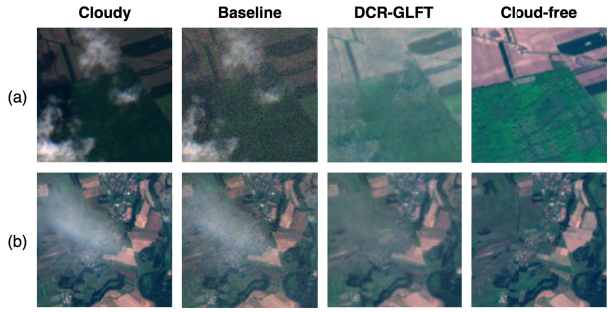


Fig. 7. Qualitative ablation study with two scenes by baseline and DCR-GLFT.

2) Density Label: To validate the effectiveness of density label, we incorporated it into the baseline to guide restoration. The baseline operates through a specialized cloud-level predictor that uses dense features to estimate cloud coverage levels. This setup facilitates targeted cloud removal strategies based on cloud density. We aim to explore whether density label can effectively enhance the network's ability to perform detailed cloud removal while maintaining the integrity of the underlying image data.

3) Density-Aware Global Context Interaction: To assess the impact of the DGCI, we designed experiments comparing it against the baseline to analyze DGCI's overall effect on model performance. The comparative results show that the DGCI block can enhance the network's capability to adapt to spatial transformations appropriately, thus improving the model's ability to restore detailed textures and structures obscured by clouds. These experimental results clearly demonstrate the role of DGCI in enhancing the accuracy of cloud-delineated areas and in maintaining the structural and textural integrity of the declouded images.

4) SFT Block: To validate the effectiveness of the SFT layer, we conducted a series of experiments by systematically incorporating or omitting the SFT layer within our network architecture. By comparing the performance of networks with and without the SFT layer, we aimed to isolate and quantify the impact of this layer on the network's ability to accurately identify and remove cloud-covered areas in satellite images. We evaluated the visual quality of the images, examining the clarity of the reconstructed areas and the preservation of textural details that are often lost under cloud cover, and analyzed the performance using quantitative metrics. The results of these experiments demonstrated that the presence of the SFT layer significantly enhances the model's spatial awareness and feature fusion capabilities. The experiments showed that the SFT layer plays a critical role in improving cloud removal performance and is practical in achieving superior image restoration outcomes.

V. CONCLUSION

In this work, we propose a novel DCR-GLFT for efficient cloud-free image reconstruction. The algorithm enhances cloud removal performance in two key aspects: first, it utilizes a shared feature approach, employing density labels as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.

guidance for cloud removal. Compared to using SAR or cloud masks as guidance, density labels offer higher operability, simplifying model input requirements and improving processing efficiency. Second, the introduction of density labels effectively controls the cloud removal process under varying levels of cloud coverage, enabling the algorithm to adaptively handle different degrees of cloud cover. This not only improves the accuracy of cloud removal but also enhances the robustness of the algorithm in diverse cloud conditions. Overall, the DCR-GLFT method demonstrates exceptional performance across various levels of cloud coverage, significantly improving the quality of cloud-free image reconstruction and showing great potential for practical applications. To sum up, this article provides a promising and controllable framework for cloud removal of optical remote sensing images, and it has high potential to be deployed for enhancing remote sensing perception task (e.g., change detection), which is one of our ongoing work on formulating a unified framework for joint cloud removal and pixel-level understanding.

REFERENCES

[1] P. K. Soni, N. Rajpal, R. Mehta, and V. K. Mishra, "Urban land cover and land use classification using multispectral sentinal-2 imagery," *Multimedia Tools Appl.*, vol. 81, no. 26, pp. 36853–36867, 2022.

[2] E. Bezerra, S. Mafalda, A. B. Alvarez, D. A. Uman-Flores, W. I. Perez-Torres, and F. Palomino-Quispe, "A cloud coverage image reconstruction approach for remote sensing of temperature and vegetation in Amazon rainforest," *Appl. Sci.*, vol. 13, no. 23, p. 12900, Dec. 2023.

[3] R. A. Frey et al., "Cloud detection with MODIS. Part I: Improvements in the MODIS cloud mask for collection 5," *J. Atmos. Ocean. Technol.*, vol. 25, no. 7, pp. 1057–1072, Jul. 2008.

[4] S. A. Ackerman, K. I. Strabala, W. P. Menzel, R. A. Frey, C. C. Moeller, and L. E. Gumley, "Discriminating clear sky from clouds with MODIS," *J. Geophys. Res., Atmos.*, vol. 103, no. D24, pp. 32141–32157, Dec. 1998.

[5] Z. Chen, P. Zhang, Y. Zhang, X. Xu, L. Ji, and H. Tang, "Thick cloud removal in multi-temporal remote sensing images via frequency spectrum-modulated tensor completion," *Remote Sens.*, vol. 15, no. 5, p. 1230, Feb. 2023.

[6] S. Xu, J. Wang, and J. Wang, "Fast thick cloud removal for multitemporal remote sensing imagery via representation coefficient total variation," *Remote Sens.*, vol. 16, no. 1, p. 152, Dec. 2023.

[7] C. Long, X. Li, Y. Jing, and H. Shen, "Bishift networks for thick cloud removal with multitemporal remote sensing images," *Int. J. Intell. Syst.*, vol. 2023, pp. 1–17, Feb. 2023.

[8] H. Pan, "Cloud removal for remote sensing imagery via spatial attention generative adversarial network," 2020, *arXiv:2009.13015*.

[9] J. D. Bermudez, P. N. Happ, D. A. B. Oliveira, and R. Q. Feitosa, "SAR to optical image synthesis for cloud removal with generative adversarial networks," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 4, no. 1, pp. 5–11, Sep. 2018.

[10] C. Grohnfeldt, M. Schmitt, and X. Zhu, "A conditional generative adversarial network to fuse SAR and multispectral optical data for cloud removal from Sentinel-2 images," in *Proc. IGARSS IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 1726–1729.

[11] A. Meraner, P. Ebel, X. X. Zhu, and M. Schmitt, "Cloud removal in Sentinel-2 imagery using a deep residual neural network and SAR-optical data fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 333–346, Aug. 2020.

[12] F. Xu et al., "GLF-CR: SAR-enhanced cloud removal with global–local fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 268–278, Oct. 2022.

[13] J. Gao, Q. Yuan, J. Li, H. Zhang, and X. Su, "Cloud removal with fusion of high resolution optical and SAR images using generative adversarial networks," *Remote Sens.*, vol. 12, no. 1, p. 191, Jan. 2020.

[14] P. Ebel, A. Meraner, M. Schmitt, and X. X. Zhu, "Multisensor data fusion for cloud removal in global and all-season Sentinel-2 imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5866–5878, Jul. 2021.

[15] M. Shi, F. Xie, Y. Zi, and J. Yin, "Cloud detection of remote sensing images by deep learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 701–704.

[16] J. Lin, T.-Z. Huang, X.-L. Zhao, Y. Chen, Q. Zhang, and Q. Yuan, "Robust thick cloud removal for multitemporal remote sensing images using coupled tensor factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5406916.

[17] F.-F. Li, H.-M. Zuo, Y.-H. Jia, Q. Wang, and J. Qiu, "Hybrid cloud detection algorithm based on intelligent scene recognition," *J. Atmos. Ocean. Technol.*, vol. 39, no. 6, pp. 837–847, Jun. 2022.

[18] X. Pan, F. Xie, Z. Jiang, and J. Yin, "Haze removal for a single remote sensing image based on deformed haze imaging model," *IEEE Signal Process. Lett.*, vol. 22, no. 10, pp. 1806–1810, Oct. 2015.

[19] Q. He, X. Sun, Z. Yan, and K. Fu, "DABNet: Deformable contextual and boundary-weighted network for cloud detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[20] J. H. Jeppesen, R. H. Jacobsen, F. Inceoglu, and T. S. Toftegaard, "A cloud detection algorithm for satellite imagery based on deep learning," *Remote Sens. Environ.*, vol. 229, pp. 247–259, May 2019.

[21] O. Hagolle, M. Huc, D. V. Pascual, and G. Dedieu, "A multi-temporal method for cloud detection, applied to FORMOSAT-2, VENUS, Landsat and Sentinel-2 images," *Remote Sens. Environ.*, vol. 114, no. 8, pp. 1747–1755, Aug. 2010.

[22] Z. Zhu and C. E. Woodcock, "Object-based cloud and cloud shadow detection in Landsat imagery," *Remote Sens. Environ.*, vol. 118, pp. 83–94, Mar. 2012.

[23] P. L. Scaramuzza, M. A. Bouchard, and J. L. Dwyer, "Development of the Landsat data continuity mission cloud-cover assessment algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 4, pp. 1140–1154, Apr. 2012.

[24] M. Le Goff, J.-Y. Tourneret, H. Wendt, M. Ortner, and M. Spigai, "Deep learning for cloud detection," in *Proc. 8th Int. Conf. Pattern Recognit. Syst. (ICPRS)*, Jul. 2017, pp. 1–6.

[25] O. R. Mitchell, E. J. Delp, and P. L. Chen, "Filtering to remove cloud cover in satellite imagery," *IEEE Trans. Geosci. Electron.*, vols. GE-15, no. 3, pp. 137–141, Jul. 1977.

[26] T. F. Chan and J. Shen, "Nontexture inpainting by curvature-driven diffusions," *J. Vis. Commun. Image Represent.*, vol. 12, no. 4, pp. 436–449, Dec. 2001.

[27] A. Maalouf, P. Carre, B. Augereau, and C. Fernandez-Maloigne, "A bandelet-based inpainting technique for clouds removal from remotely sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 7, pp. 2363–2371, Jul. 2009.

[28] P. Singh and N. Komodakis, "Cloud-GAN: Cloud removal for Sentinel-2 imagery using a cyclic consistent generative adversarial networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 1772–1775.

[29] X. Wen, Z. Pan, Y. Hu, and J. Liu, "Generative adversarial learning in YUV color space for thin cloud removal on satellite imagery," *Remote Sens.*, vol. 13, no. 6, p. 1079, Mar. 2021.

[30] Y. Zi, F. Xie, X. Song, Z. Jiang, and H. Zhang, "Thin cloud removal for remote sensing images using a Physical-Model-Based CycleGAN with unpaired data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] G. Scarpa, M. Gargiulo, A. Mazza, and R. Gaetano, "A CNN-based fusion method for feature extraction from sentinel data," *Remote Sens.*, vol. 10, no. 2, p. 236, Feb. 2018.

[32] H. Shen, J. Wu, Q. Cheng, M. Aihemaiti, C. Zhang, and Z. Li, "A spatiotemporal fusion based cloud removal method for remote sensing images with land cover changes," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 3, pp. 862–874, Mar. 2019, doi: 10.1109/JSTARS.2019.2898348.

[33] Q. Zhang, Q. Yuan, Z. Li, F. Sun, and L. Zhang, "Combined deep prior with low-rank tensor SVD for thick cloud removal in multitemporal images," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 161–173, Jul. 2021.

[34] J. Gao, Y. Yi, T. Wei, and G. Zhang, "Sentinel-2 cloud removal considering ground changes by fusing multitemporal SAR and optical images," *Remote Sens.*, vol. 13, no. 19, p. 3998, Oct. 2021.

[35] P. Ebel, Y. Xu, M. Schmitt, and X. X. Zhu, "SEN12MS-CR-TS: A remote-sensing data set for multimodal multitemporal cloud removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.

[36] A. Levin, D. Lischinski, and Y. Weiss, "A closed-form solution to natural image matting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 30, no. 2, pp. 228–242, Feb. 2008.
[37] M. Mirza and S. Osindero, "Conditional generative adversarial nets," 2014, *arXiv:1411.1784*.
[38] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 1125–1134.
[39] A. Ramjyothi and S. Goswami, "Cloud and fog removal from satellite images using generative adversarial networks (GANs)," 2021.
[40] H. Emami, M. M. Aliabadi, M. Dong, and R. B. Chinnam, "Spa-GAN: Spatial attention GAN for image-to-image translation," *IEEE Trans. Multimedia*, vol. 23, pp. 391–401, 2020.
[41] X. Wen, Z. Pan, Y. Hu, and J. Liu, "An effective network integrating residual learning and channel attention mechanism for thin cloud removal," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[42] D. Ma, R. Wu, D. Xiao, and B. Sui, "Cloud removal from satellite images using a deep learning model with the cloud-matting method," *Remote Sens.*, vol. 15, no. 4, p. 904, Feb. 2023.
[43] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.
[44] P. Ebel, V. S. F. Garnot, M. Schmitt, J. D. Wegner, and X. X. Zhu, "UnCRtainTS: Uncertainty quantification for cloud removal in optical satellite time series," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 2086–2096.
[45] T. Wang, X. Yang, K. Xu, S. Chen, Q. Zhang, and R. W. H. Lau, "Spatial attentive single-image deraining with a high quality real rain dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12270–12279.
[46] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 136–144.



Quan Rui is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China.

His research interests include computer vision and remote sensing image processing.



Shiyuan He is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China.

His major research interests include multimedia, computer vision, and machine learning.



Tianyu Li received the Ph.D. degree from the School of Electrical and Computer Engineering, Purdue University, West Lafayette, IN, USA, in 2023.

He is currently a Post-Doctoral Researcher with the Center for Future Media, University of Electronic Science and Technology of China, Chengdu, China. His research focuses on image processing using statistical models and deep learning models, as well as time-series forecasting and anomaly detection in time series.



Guoqing Wang (Member, IEEE) received the Ph.D. degree from The University of New South Wales, Sydney, NSW, Australia, in 2021.

He is currently with the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China. He has authored or co-authored more than 40 scientific articles at top venues, including *International Journal of Computer Vision* (IJCV), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY (TIFS), ICCV, and ACM MM. His research interests include machine learning and unmanned systems, with special emphasis on visual perception and embodied agents.

Dr. Wang's research work at UNSW has been recognized with Australian Dean's Award for Outstanding Ph.D. Thesis.



Ningjuan Ruan received the Ph.D. degree in electronic science and technology from Nanjing University, Nanjing, China, in March 2019.

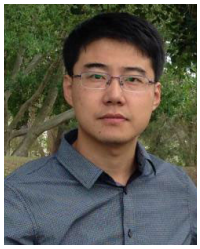
She is currently the Director of the Advanced Optical Remote Sensing Beijing Key Laboratory. She is also affiliated with Beijing Institute of Space Mechanics and Electricity, Beijing, China.

Dr. Ruan is also a member of the Expert Committee of the Core Technology Integration Platform at Nanjing University, under the Ministry of Education, and the Academic Committee of the Key Laboratory of Space Optoelectronic Detection and Perception, Ministry of Industry and Information Technology.



Lin Mei received the Ph.D. degree from Xi'an Jiaotong University, Xi'an, China, in 2000.

He is currently a Researcher with the Donghai Laboratory, Zhejiang, China. His research interests include computer vision, artificial intelligence, the Internet of Things, and big data.



Yang Yang (Senior Member, IEEE) received the bachelor's degree from Jilin University, Changchun, China, in 2006, the master's degree from Peking University, Beijing, China, in 2009, and the Ph.D. degree from The University of Queensland, Australia, in 2012, under the supervision of Prof. H. T. Shen and Prof. X. Zhou.

From 2012 to 2014, he was a Research Fellow under the supervision of Prof. T.-S. Chua with the National University of Singapore, Singapore. He is currently with the University of Electronic Science and Technology of China, Chengdu, China.



Heng Tao Shen (Fellow, IEEE) received the B.Sc. (Hons.) and Ph.D. degrees from the Department of Computer Science, National University of Singapore, Singapore, in 2000 and 2004, respectively.

His research interests include multimedia search, computer vision, artificial intelligence, and big data management.

Prof. Shen is a member of Academia Europaea and a fellow of ACM and OSA.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:35 UTC from IEEE Xplore. Restrictions apply.